

James Brown

Gardener of emergent systems — human and machine.

I've come to believe good systems aren't built so much as tended. For twenty-odd years I've been lucky to work on **living systems — distributed data, autonomous teams, and lately AI agents** — usually alongside people sharper than me. You don't force a system to change; you try to be a useful catalyst, and let it find its own shape. Mostly, my craft is **leading and scaling the teams** — and lately the agents — that do the building: a force-multiplier more than a lone builder.

↩ Twenty years building high-throughput systems — and the teams behind them. From a sports-betting startup's rise to a **\$1.5B float**, honed at **ThoughtWorks**, to steering AI & agentic engineering at a **\$1-trillion** asset manager. The story's below; there's a little lab at the end.



↑ **How I actually lead** — my Leadership OS, in depth ↗

\$1.5B
NYSE float • where I grew up

\$1T
AUM • where I am now

- [LinkedIn](#)
- [GitHub](#)
- [Medium](#)
- [Email](#)

THE JOURNEY • 2006 → NOW • READ TOP TO BOTTOM

One continuous thread — from a support desk to a \$1-trillion bank.

Each role forged a trait the next inherited. Read top to bottom — it's how I got from there to here.

- **main** • the through-line — read top to bottom (`git log --reverse`)

`tag: refs/edu/cardiff` (root commit)
BSc Geology & Earth Science • Cardiff University • 2001-04 — the first seams I ever learned to read.
+ Windsurfing Club

Betfair • Market Analyst • where the bug bit
Where it started — cutting my teeth on the live trading platform during in-play sports, opening and closing markets in real time as bets traded against each other.

Betgenius → Genius Sports • 2006 → 2019 `feat/genius-sports` TAG: V-FLOAT-1.5B

Genius Sports — a warehouse startup to a \$1.5B float

Tech Support > QA > Developer > Senior Dev > **Team Lead**

B2B support desk to team lead • Aug 2006 → Feb 2019
100k+ msg/s • sub-second→nanosecond latency • 30 → 1,000+ people
I was lucky to join a 30-person warehouse startup in Kentish Town — starting on the support desk — and over twelve years worked my way up to team lead as it grew into a **\$1.5B NYSE-listed company**. Alongside some brilliant engineers we built distributed, event-driven systems that "herd data from low value to high value" — most still run today, powering well-known sports-betting platforms. It's where I learned to design for **the machine and the human at once**.

FORGED *mechanical + human sympathy — build for whoever is doing the work: us, or the metal.* ↗ [Sympathy](#)
"Quality is the only thing that allows for sustainable speed." ↗ [Leadership OS](#)

► **Talk** • Continuous Visibility, No More Dashboards (JFokus)

[commit on feat/genius-sports](#)

[TAG: MSC-DISTINCTION](#)

MSc Computer Software Engineering · Distinction · University of Westminster · 2008–10 — studied while working; branched off the day job and merged a distinction back.

[/thoughtworks](#)

[feat/thoughtworks](#)

Lead Software Development Consultant · 2019–2022 · client-facing as **Principal Engineer** on the Client Leadership Team (CLT)

the firm where Continuous Delivery, Data Mesh, the Tech Radar & the Architecture Advice Process were born — the engineering-excellence pedigree behind the three chapters below. I was also lucky to speak on conference stages throughout these years: Sweden, Norway, Morocco, London & Milan.

ThoughtWorks · Engineer

[gig/home-office](#)

[TAG: SYSTEM-REGISTER](#)

Hands-on team lead — UK Home Office

system-register · Quarkus/Java + React · **elite DORA**

With a sharp team we delivered government-grade software to **elite DORA standards** — security-conscious continuous delivery, **open-sourced** for the public, and work that went on to surface findings that reached national headlines. Polyglot by habit — Java, Kotlin, Python, Rust, Terraform, React, Vue — and quietly obsessive about quality engineering: TDD, pairing, trunk-based development.

FORGED *government-grade rigour as the default, not the ceremony.*

"A demo beats a deck." · A systems thinker's TDD — reframed as math: quality is a matter of time, not opinion. [↗ To TDD?](#)

[</> Open-source code](#)

ThoughtWorks · Contributor

[gig/bmmro](#)

[FOUNDER · 0→1](#)

BMMRO — an idea nobody asked for

£1M+ of free technology · ~12 contributors · 4 iterations & counting

Nobody briefed it — I just believed marine scientists deserved better tools. I pitched it, and was humbled when people stepped off the bench to build it after a few homemade videos and posters. Together they made something that's run for **years**, quietly changing how Bahamas researchers work. I never wrote a line of its code; my part was helping form the **team that could** — then trusting them with it.

FORGED *0→1 initiative — the nerve to start something, and the sense to hand it over.*

"A concept is more likely to starve in isolation than be stolen. Speak up, share, write, talk."

[↗ Leadership OS](#)

[▶ Watch the ThoughtWorks film](#)

[</> See the code](#)

ThoughtWorks · Leader

[gig/sims](#)

[TAG: SIMS-NEXT-GEN](#)

Head of Technology — SIMS (ParentPay Group)

5 products in 18 months · 200 devs · 7 DORA-elite teams · real-time Data Mesh

Europe's premier ed-tech, with a **75% market share** that was slipping 10%/year and **no new product in 8 years**. I was trusted to lead the turnaround — the reorg, the architecture, the engineering-practice shift — and with a genuinely talented team we shipped **SIMS Next Gen**, the cloud-native rebuild, then four more products in 18 months on a real-time Data-Mesh platform the competition's batch systems couldn't match. I left behind **seven autonomous, product-led teams** at elite DORA — easily the thing I'm proudest of.

FORGED *org design as the first act of architecture — Conway's Law, cultivated deliberately.*

"If more than five people start using it, sound the alarm — the idea has proven its value." [↗ Citizen Devs](#)

[↗ SIMS Next Gen](#)

[▶ Talk · Big Estimates with Graph Theory](#)

+ Plenty more across these years — including a technical RFP for a **3-million-trades/second** platform (Kotlin · Kafka · hexagonal architecture).

The City · Sep 2022 → present

[HEAD → main](#)

[* \(CURRENT\)](#)

Schroders — Enterprise Engineering Lead

Staff engineer · 200-year-old asset manager · **\$1 trillion AUM** · **10× engagement** on internal products

[Agentic enablement](#)

[Skills marketplace](#)

[Team-metrics products](#)

[Mobile app delivery](#)

[Vision & strategy · global](#)

These days I'm grateful for the chance to work across several programmes at once — a skills marketplace, team-metrics products, mobile delivery — all with **tracked, 10×-growing engagement**, while helping shape engineering **vision and strategy at a global level**. I help steer the **agentic strategy**: carefully rolling out autonomous agents for critical, audited, regulated scrutiny work. [↗ Substrate](#) Mostly I try to be the heat shield — turning technical reality into business risk so the people building can stay focused.

TURNING *the leadership heat-shield, and change as immunology — win the culture or lose the transformation.*

"A leader moves like water to do whatever is needed to progress the idea — to success or failure."

[▶ AI expert panel · XT26](#)

[↗ Speed vs Quality · NSTC talk + essay](#)

\$ git checkout **play/speciate** — a detached branch I keep for fun, never merged to main. It's just below ↓

And for fun, I point the craft at toy problems.

Twenty years of systems habits don't switch off at the weekend. **speciate** is a personal sandbox — an artificial-life world where DNA-driven creatures flock and evolve. Really it's an excuse to aim low-latency, trading-platform thinking at a game, and to see how far AI agents can build genuine systems code. A technical showcase, not a product.

speciate // a side project • code 100% agent-authored

github.com/JBJamesBrownJB/speciate ↗

900K

creatures @ 20Hz • 0 stall frames

0.77%

branch-miss • perf_event counters

σ 0.8ms

frame-time deviation

IPC 4.25

movement hot-loop

761

files • 124 Rust modules

6.3×

Rayon parallel speedup

Twenty years of trading-platform habits, in Rust. A data-oriented ECS core — Struct-of-Arrays archetypes, a contiguous spatial buffer packed by a zero-allocation parallel counting sort — keeps the hot working set cache-resident. **"My L3 is my RAM."** Movement runs on Rust's fearless parallelism (Rayon) for a **6.3× speedup**, confirmed by `perf_event` counters: IPC 4.25, 0.77% branch-miss.

Joined to the richest frontend ecosystem in the world. That Rust core meets JavaScript — PixiJS / WebGL — at a **zero-copy seam** where most hybrids die: positions cross the boundary as a shared `Float32Array` over a NAPI-RS double buffer, no per-frame serialization tax. Two worlds, mechanical sympathy intact.

And built almost entirely by AI agents — an auditable agentic SDLC (named subagents, sprint commands, TDD-enforcing hooks, AGENTS.md contracts), the same methods now shaping the agentic strategy at Schroders.

git shortlog -sn — who built it

claude (agents) +152

JB (human) +40

The human set the genome; the agents grew the colony.
A decade after BMMRO — same 0→1 instinct, now with machines.

— LET'S CULTIVATE SOMETHING THAT SHIPS

James Brown

● LinkedIn

● GitHub

● Medium

● Email



Contributor to the thinking around the **Architecture Advice Process** — Andrew Harmel-Law, **Facilitating Software Architecture** (O'Reilly). [On Amazon](#) ↗

A timeline that builds the person.
The hero is a **live emergent simulation** — it speciates into the engine at the end.
Flocking after Craig Reynolds' Boids (1986) — three local rules, emergent order.
Single source → HTML + dense print PDF.
Co-Authored-By: Claude Opus 4.8 (1M context) — this CV, like speciate, was grown with agents.